Evaluation of Trivial Dialogue Phrase Databases Through Practical Application to User-Computer Conversation - Case Study: English - Spanish

Calkin S. Montero and Kenji Araki

Graduate School of Information Science and Technology, Hokkaido University,
Kita 14-jo Nishi 9-chome, Kita-ku, Sapporo, 060-0814 Japan
{calkin,araki}@media.eng.hokudai.ac.jp

Abstract. This paper presents the results obtained during the application of automatically generated open-domain trivial dialogue phrase databases to conversational partners. The databases were generated by a language independent algorithm. Evaluative experiments were carried out for English and Spanish. Results show the applicability of the analyzed phrase databases.

Keywords: User-computer conversation, trivial dialogue phrases, dialogue evaluation.

1 Introduction

The challenging goal of implementing computer applications capable to sustain a conversation with humans has long been of particular interest to NLP-researchers. The significant advances in the core fields of human language technology, i.e. computational linguistics, speech technology, etc., have led to the recent development of dialogue systems applications. These applications range from goal specific spoken dialogue systems, e.g., Jupiter, providing a telephone-based conversational interface for international weather [1], airline travel information systems [2], restaurant guides [3], telephone interfaces to emails or calendars [4], car navigation systems [5], and so forth, to attempts to simulate human trivial dialogue - chat - e.g. ELIZA [6].

In this respect, natural language generation (NLG) as a sub-field of computational linguistics and language-oriented Artificial Intelligence (AI) has devoted itself to researching and simulating the production of written or spoken discourse. The applications and limitations of language generation have been widely studied. In this regard well-known approaches are the *canned text* approach, in which the computer prints out a text given by a programmer, and the *template filling* approach, in which predetermined templates are filled up to produce a desired output. Other approaches include the *phrase based* approach, which can be conceptualized as generalized templates that recursively match parts of the input, and the *feature based* approach, in which the generation can be conceptualized as a collection of features appropriate for each part of the input [7].

B. Apolloni et al. (Eds.): KES 2007/ WIRN 2007, Part II, LNAI 4693, pp. 361–368, 2007.
© Springer-Verlag Berlin Heidelberg 2007

The template filling approach has been applied to the simulation of human chat as a language generation technique. One of the most famous examples of conversational systems, ELIZA [6], uses the template filling approach to generate the system's response to a user input, however due to the templates limitations the dialogue tends to become repetitive and monotonous, hardly ever satisfying the user's expectations.

In more recent research Inui et al. [8] have used a corpus based approach to language generation for dialogue system. In their approach, Inui et al. (2003), applied keyword matching in order to extract sample dialogues from a *dialogue corpus*, i.e., utterance-response pairs. However the overall evaluation of the system is not given, making it hard to compare the system's performance. Other conversational systems, e.g. ALICE by Wallace [9] and Jabberwacky by Carpenter [10], apply as well a corpus based approach to NLG in order to retrieve system's responses to a user inputs being evaluated most of the time by on-line users. A disadvantage of corpus based conversational system is that they suffer from the tedious labor of handcrafting their dialogue corpus[1]. In previous research we have proposed a language independent approach for the automatic generation of trivial phrases database, arguing to reduce the load of work and the amount of time needed in order to create a chatbot dialogue corpus [11].

However, it is difficult to objectively evaluate the performance of a conversational partner - chatbot - system. One reason is that there are not fixed tasks to accomplish throughout the dialogue. Another reason is that topical coherence must prevail in spite of the topic-shifts that occur during the conversation [12]. Therefore the evaluation of a chatbot is in general solely based on the user's satisfaction. This can be observed during chatbot evaluative contests, e.g., the Loebner Prize competition [13], where the system is evaluated based on the judges opinion on how well the system responds to a set of fixed questions.

In this paper it is described the evaluation of the applicability of automatically generated trivial dialogue phrases[2] databases to chatbot programs. We apply the semantic differential (SD) method in order to identify users attitude towards the chatbots *overall performance* during the dialogue. Results show the applicability of the analyzed phrase databases.

2 System Overview

In this system the automatically generated trivial dialogue phrases database is used as part of a chatbot database. The system general overview is shown in Fig.1. A brief explanation of the creation of the Phrases DB is given in Sect.2.1.

[1] ALICE chatbot database (ALICE brain) took over 10 years work to accomplish. The Jabberwacky database is being developed since 1988
http://www.alicebot.org/superbot.html
http://alicebot.org/articles/wallace/dont.html
http://feeling.jabberwacky.com:8081/j2about

[2] A trivial dialogue phrase is referred as an expression used by a chatbot program as the answer of a user input.

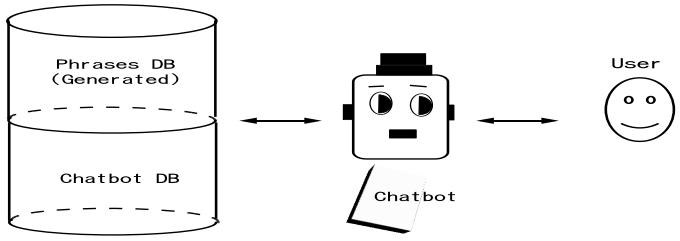


Fig. 1. System Overview

2.1 Generated Trivial Phrases DB

The generation of the trivial Phrases DB is performed using a GA-like transfer approach, details given in [14]. Fig.2 shows the generation process. A brief description of each step is given hereunder.

Initial Population. In the population selection process a small population of phrases are selected randomly from a set of phrases given to the system beforehand. For the experiments, this set of phrases consisted of 30 phrases for English and 20 phrases for Spanish.

Crossover. In order to ensure a language independent method the algorithm does not use syntactic information (part of speech tagging). Therefore, in order to avoid the distortion of the selected phrase during the initial population process due to completely blind crossover, the crossover rate was set to 0%.

Mutation. During the mutation process, each one of the seed phrases selected in the initial population is mutated at a rate of $1/N$, where N is the total number of words in the phrase. The mutation is performed through a transfer process, using features randomly selected, i.e., nouns, adjectives, adverbs, given beforehand.

Evaluation. In order to evaluate the newly generated expression, the WWW is used as knowledge base. Since matching an entire phrase on the Web might result in very low retrieval, or in some cases non retrieval at all, we apply the sectioning of the phrase into its respective n-grams. First, each n-gram for a given phrase is evaluated according to its frequency, i.e, number of hits, on the WWW. Then the system evaluates the phrase as "good" if all of its n-grams were labeled "accepted" by the system, that is to say, all of the n-grams are above the given threshold. If for a given phrase there is at most *one rejected n-gram* or *one weakly accepted n-gram*, the n-gram is below a given threshold, and the phrase is evaluated as "usable", otherwise the phrase is "rejected".

In order to verify its portability, the algorithm was applied to the generation of English and Spanish trivial dialogue phrases. According to a user evaluation, 243 out of the 1,767 generated phrases for Spanish were *semantically useful* for dialogue. In the case of English, 1,030 out of 2,512 phrases generated were

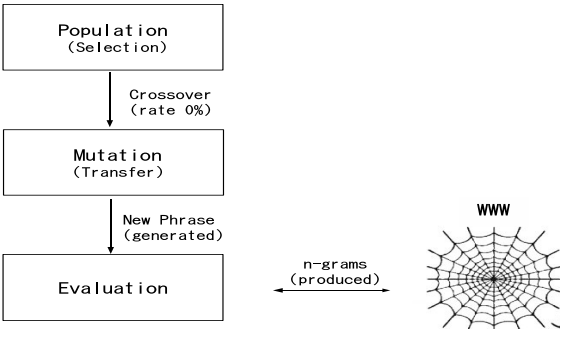


Fig. 2. Phrases DB Generation Process

semantically useful[3]. In this paper, these English and Spanish semantically useful phrases were applied as part of a chatbot database as explained in Sect.3.

3 Design Experiments: Databases Evaluation and Results

As stated before, it is difficult to empirically evaluate a conversational partner system. For the experiments a comparative evaluation between a plain system (baseline) and a system using the generated database was performed. Phrases from both English and Spanish databases were evaluated within real dialogue context. The experiment combined aspect for the evaluation of the chatbots performance based on the *semantic differential (SD) method* measures, as well as other criteria such as user satisfaction and acceptance.

The SD method was developed by the psychologist and communication scholar Charles E. Osgood [15]. This method connects scaled measurement of attitudes with the *connotative meaning* of words, i.e., subjective cultural or emotional association with a word. Osgood worked with the semantics of words and ideas involved in scaling opinions and created the SD method to plot a psychological distance between words by mapping a subject's connotations of the words. During the experiment this method was applied to determine the user's attitude towards the chatbots performance.

In the SD method there are three major factors or dimensions of judgment:

- Evaluative factor (good - bad)
- Potency factor (strong - weak)
- Activity factor (active - passive)

For the experiments, 15 adjective-pairs were used within a 7 points ranking scale where "7" was the strongest, positive meaning, and "1" was the weakest,

[3] The term *"semantically useful"* refers here to the ability of the phrase to convey information within a given context. This implies that a given phrase does not necessarily have to be grammatically correct in order to be understood and used.

negative meaning. Fig.3 shows the selected adjective-pairs[4]. Here the *evaluative factor* is represented by the pairs {a-e, g, i, k, l, n}, the *potency factor* is represented by {j}, and the *activity factor* is represented by {f, h, m, o}.

a. Fun	7	6	5	4	3	2	1	boring
b. Interesting	7	6	5	4	3	2	1	uninteresting
c. Human-like	7	6	5	4	3	2	1	un-natural
d. Good	7	6	5	4	3	2	1	bad
e. Entertaining	7	6	5	4	3	2	1	unentertaining
f. Communicative	7	6	5	4	3	2	1	unexpressive
g. Understandable	7	6	5	4	3	2	1	not-understood
h. Innovative	7	6	5	4	3	2	1	monotonous
i. Intelligent	7	6	5	4	3	2	1	dumb
j. Easy	7	6	5	4	3	2	1	difficult
k. Enjoyable	7	6	5	4	3	2	1	unpleasant
l. Kind	7	6	5	4	3	2	1	unkind
m. Relaxing	7	6	5	4	3	2	1	stressful
n. Like	7	6	5	4	3	2	1	dislike
o. Positive impression	7	6	5	4	3	2	1	negative imp.

Fig. 3. Adjective-pairs Seven-points Scale Ranking

The performed experiments were carried out as follows. The baseline system, ALICE[5] [9], uses Artificial Intelligence Markup Language (AIML), a version of XML, to represent patterns (user's input) and templates (chatbot's response). In a preliminary stage, the databases (English and Spanish correspondingly) were converted into AIML files of general patterns to be used externally as part of the chatbot inner database. However, users feedback (3 non native English speakers, 2 of them native Spanish speakers) showed that when the chatbot used the databases, its performance (in terms of user satisfaction) was much lower than the chatbot performance without the databases. A fundamental cause is considered to be the lack of correlation between the user input and the context for using the phrases from the databases as a response. Therefore, the phrases from the generated databases (English and Spanish) were inserted manually into the chatbots inner databases. In this way it is possible to evaluate the quality of the phrases within the dialogue as well as how influential these phrases were to the chatbots overall performance. Three native Spanish speakers and three non native English speakers (university students, 4 males, 2 females) were prompted to have a conversation with the baseline chatbot and the chatbot using the augmented databases for both Spanish and English. The results of the averaged evaluation of the chatbots performance according to the SD method are shown in Fig. 4 for Spanish and Fig. 5 for English.

[4] Adjectives were selected according to their appearance in the Web after searching for *"chatbot is"*.

[5] Public version of 2001.

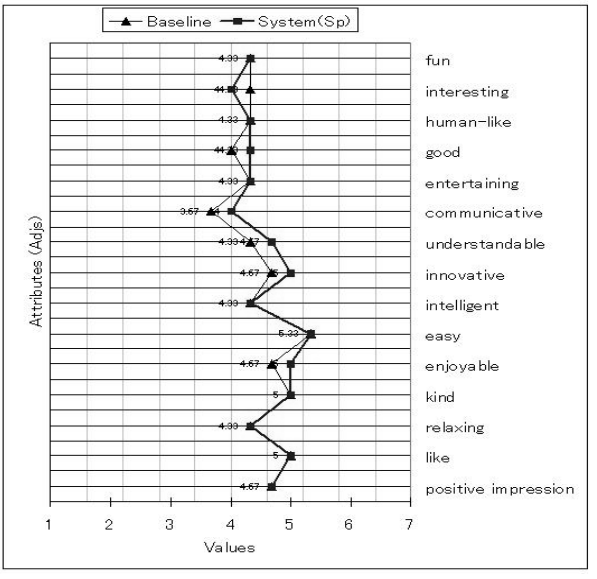


Fig. 4. Average Evaluation for Spanish System

In Fig.4 it is shown that the baseline system was evaluated as more "interesting" than the system using the generated Spanish database (System(Sp)). However, looking at the evaluation for "good" the System(Sp) is evaluated higher that the baseline system. This apparent contradiction could be explained regarding to the users *novelty feeling* towards the chatbots. The users were prompted to have a chat with the baseline system *first*. When they were prompted to have a chat with the System(Sp), the novelty feeling seemed to decrease since they had already used a similar system. However, observing the performance itself, the System(Sp) scored higher than the baseline system, showing improvement.

In Fig.5 it is shown that the baseline system was evaluated as more "understandable" than the system using the generated English database (System(Eng)). This might be considered caused by the used of *semantically usable phrases*, which does not imply that the phrases are always *completely grammatically correct*. However, with the use of phrases from the automatically generated database the System(Eng) outperformed the baseline system. The results shown in these figures reveal the validity of the application of the automatically generated databases to conversational systems.

3.1 Discussion

When observing the experimental results, the English database appears to work better than the Spanish one. These results are presumably caused by the difference in size of the databases. However, the general tendency of the systems using the databases is to outperform the baseline ones.

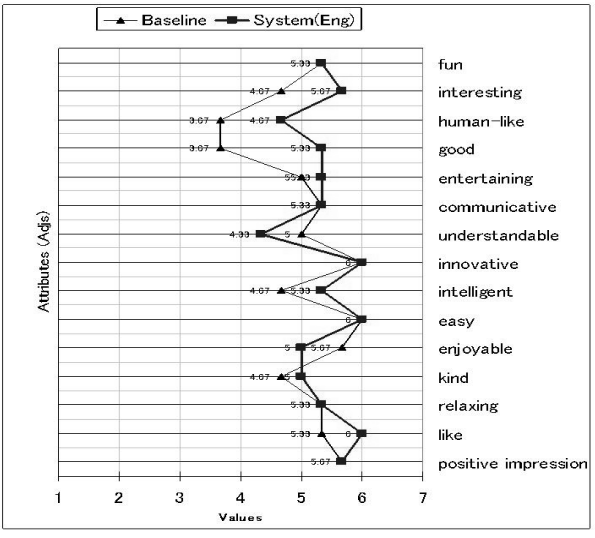


Fig. 5. Average Evaluation for English System

The results also suggest that the placement of the new databases was an important factor that showed to influence the performance of the chatbots. When the phrases of the databases were used to match general patterns, the performance of the systems was considerably lower than the performance of the systems without the phrases. On the other hand, when the databases were inserted to match specific patterns the systems performance showed improvement. Determining the appropriate time within a given context when a phrase might be used it is still a topic that requires deep research.

4 Conclusion and Future Work

In this paper the practical evaluation of automatically generated trivial dialogue phrase databases was carried out. The databases were evaluated for Spanish and English in conversational context using the SD method. Results showed a betterment in the system performance when the databases were applied, indicating the validity of the databases for conversational systems. As an ongoing work there is room for researching on the applicability of the conversational systems to interactive question-answering (QA).

Acknowledgment

This work has been partially supported by a grant of the Ministry of Internal Affairs and Communications Strategic Information and Communications R&D Promotion Program (SCOPE).

References

1. Zue, V., Seneff, S., Glass, J., Polifroni, J., Pao, C., Hansen, T., Hetherington, L.: Jupiter: A telephone-based conversational interface for weather information. IEEE Transactions on Speech and Audio Processing 8, 85–96 (2000)
2. Bratt, H., Dowding, J., Hunicke-Smith, K.: The SRI telephone-based ATIS system. In: Proceedings of the ARPA Spoken Language System Technology Workshop, pp. 22–25 (1995)
3. Lau, R., Flammia, G., Pao, C., Zue, V.: Webgalaxy: Beyond point and click - a conversational interface to a browser. In: Proceedings of the 6th International WWW Conference, pp. 119–128 (1997)
4. University of Texas at Austin: Smartvoice, `http://www.utexas.edu/its/smartvoice/`
5. Pellom, B., Ward, W., Hansen, J., Cole, R., Hacioglu, K., Zhang, J., Yu, X., Pradhan, S.: University of Colorado dialog systems for travel and navigation. In: Human Language Technology Conference (2001)
6. Weizenbaum, J.: Eliza a computer program for the study of natural language communication between man and machine. Commun. ACM 9(1), 36–45 (1966)
7. Web file: Natural language generation, `http://www.dfki.de/fluids/Natural_Language_Generation.html`
8. Inui, N., Koiso, T., Nakamura, J., Kotani, Y.: Fully corpus-based natural language dialogue system. In: Natural Language Generation in Spoken and Written Dialogue, AAAI Spring Symposium (2003)
9. Wallace, R.: A.L.I.C.E. artificial intelligence foundation (2005), `http://www.alicebot.org`
10. Carpenter, R.: Jabberwacky. learning artificial intelligence, `http://www.jabberwacky.com` `http://www.icogno.com/`
11. Montero, C.S., Araki, K.: Is it correct? - towards web-based evaluation of automatic natural language phrase generation. In: Proceedings of the COLING/ACL 2006, Interactive Presentation Sessions, pp. 5–8 (2006)
12. Herring, S.C.: Interactional coherence in CMC. Journal of Computer Mediated Communication 4(1) (1999)
13. Web file: Loebner Prize in artificial intelligence, `http://www.loebner.net/Prizef/loebner-prize.html`
14. Montero, C.S., Araki, K.: Unsupervised language independent genetic algorithm approach to trivial dialogue phrase generation and evaluation. In: NLDB07 - 12th International Conference on Applications of Natural Language to Information Systems (2007)
15. Osgood, C., Suci, G., Tannenbaum, P.: The Measurement of Meaning. University of Illinois Press, Urbana (1957)